Final Report

Section I - Introduction

1.

Problem: No one wants to toss the frisbee with me. Solution: Frisbee launching device.

2.

User: Members of our group and other Ultimate Frisbee/Frisbee Golf players.

3.

Requirements:

- Frisbee clean flight

Target: Disc has sufficient spin and launch velocity to maintain stable flight without wobble.
Priority: Constraint
Verification: Visual observation based on known characteristics of stable frisbee flight.

- Reliable frisbee loading

Target: ≥95% successful loading rate without clogging or jamming.
Priority: Constraint
Verification: Measure success rate through prototype testing.

- Launch distance

Target: At least 15 m launch distance.
Priority: Objective
Verification: Measure horizontal distance traveled by the frisbee.

- Adjustable throw types

Target: Three distinct throw profiles (Outside-In, Inside-Out, Flat).
Priority: Objective
Verification: Measure and compare lateral flight path deviation during flight and at landing.

- Adjustable launch distance

Target: At least two distinct distance settings (short and long range).
Priority: Objective
Verification: Measure launch distances and confirm they fall within the intended ranges.

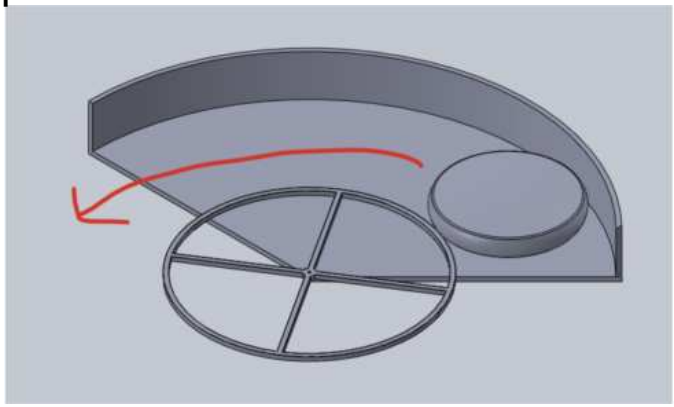
Section II - Prototypes



Prototype 1:

1.

Prototype:



2.

Testing:

a.Risky Unknown: curved vs straight track, possible problems with flywheel frisbee interplay
b. Hypothesis: The curved track will perform better because the wheel is in contact wih the frisbee for a longer period of time so more energy is transferred into the frisbee
c.Created rough cardboard models of frisbee catching structure, as well as rough curved and straight track prototypes tested with hand spun cardboard flywheel. We measured the success rate of the catching structure, as well as visually evaluated the curved and straight track designs. A catching structure prototype that worked about 80% of the time was good enough for proof of concept. The interaction between the flywheel and the frisbee for the two track designs, ie how well the flywheel comes into contact with the frisbee, and which seems to impart more kinetic energy to the frisbee.

3.

i. What did we learn:

-The curved track is superior because the wheel stays in the contact with the frisbee longer
-We will use a circular track around the flywheel to propel the frisbee rather than a straight track like a baseball launcher.

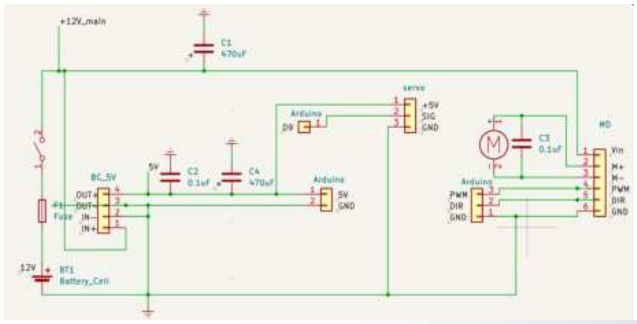
ii. Root Cause Analysis:

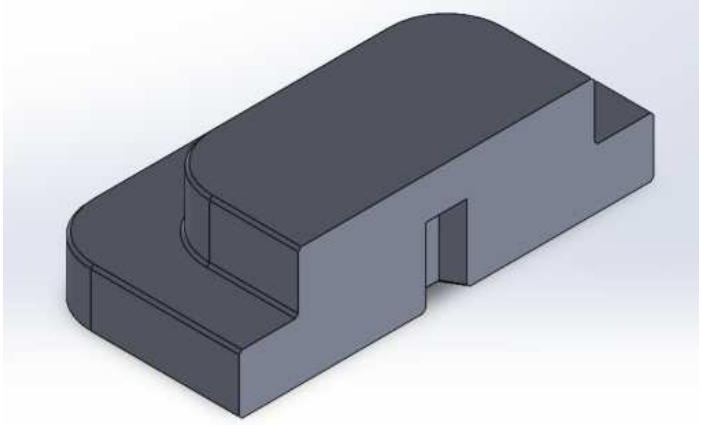
A. Flywheel slips against frisbee
B. Not enough friction between them, lack of grip or normal force
C. Ensure frisbee is snugly fit between track and flywheel, and cover the outer edge of the flywheel in a grippy material with some give

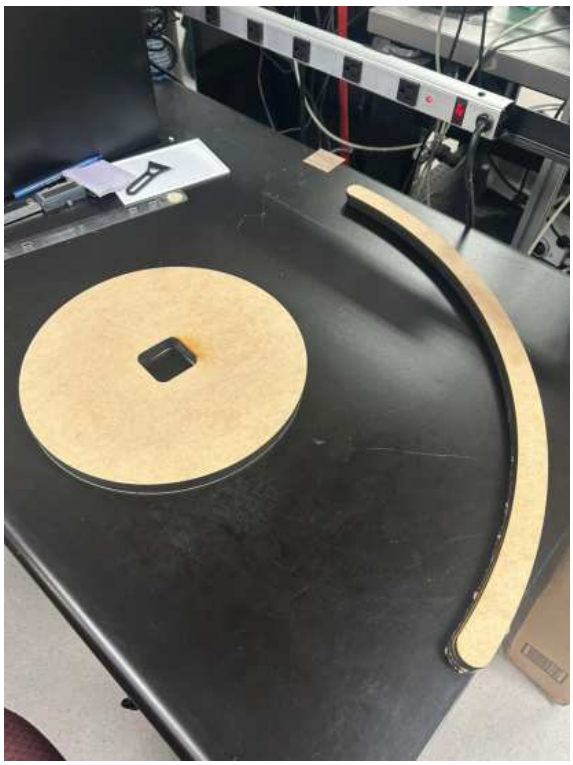
Prototype 2:

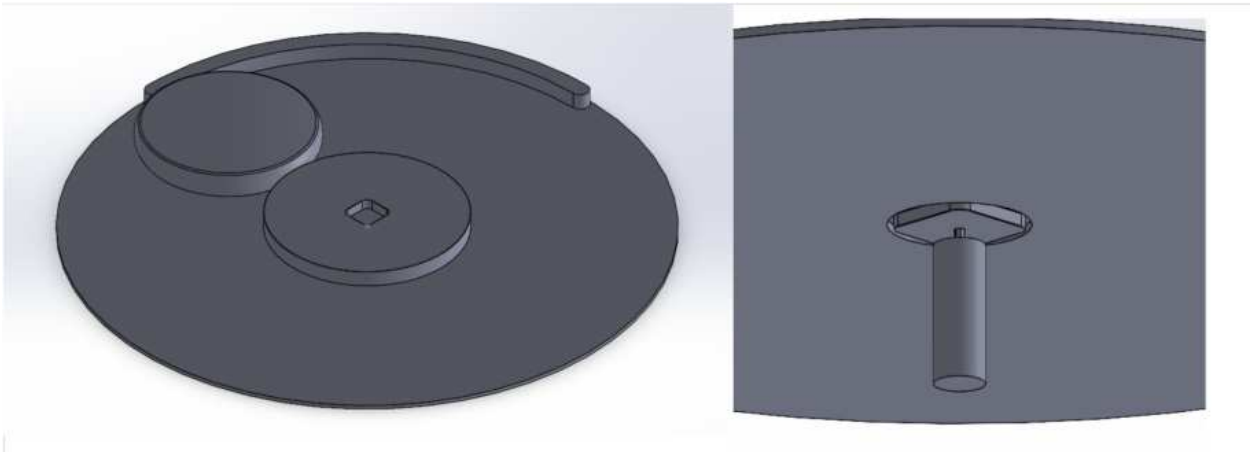
1.

Prototype:









2.

Testing:

In order to test the effectiveness of the flywheel we need a motor with enough torque/rpm, the necessary power supply and controller for the motor, and the track/wall that the frisbee glides across as it is launched. Once our prototype is assembled, we can perform tests launching frisbees to see how far they go, and based on the results of the tests, we can adjust the flywheel to better suit our needs. Most likely adjusting the weight distribution is the easiest way to change the inertia without a complete redesign. We are measuring the distance that the frisbee travels after being launched by the device. (Procedure only has motor arrived late)

3.

i. Calculations: Based on the following inputs:

Frisbee exit velocity = 15 m/s
R_flywheel = 15 cm
4 x ¼" layers
Initial spinup time = 20 seconds

The following results are given:

Flywheel mass = 1.25 kg
Frisbee RPM = 1060
Motor Torque = 0.07 Nm
RPM % Drop per shot = 20 %

ii. This confirms we can create an exit speed and RPM that aligns with initial specifications with the motor that we ordered.

Prototype 3:

1.

Prototype:





2.

Testing:

We assembled our entire device with the full circuit and track. After doing this we discovered that the disk will absolutely launch when it contacts the flywheel and our motor is adequately specced but a few of our other systems are a bit off. For example the motor shaft mount is too high off of the bottom of the track so the frisbee does not make good contact with the wheel. We noticed that the support legs were not very stable and could not hold the track with the motor spinning. We also noticed that the servo would slide over the top of the frisbee when pushing it but we did determine that was an issue with the paddle not the servo itself.

3.

i.

We tested, the motor shaft mount, the motor, the servo, the track, and the loading mechanism, and the angle adjustment system

Why is this important?

Because of our testing we have determined that the most crucial things are what we need to work on. (Shaft Mount, loading mechanism, angle adjustment)

ii.

RCA1:

Root Problem: The motor shaft mount was too shallow and the flywheel sits about one inch above the track where we want it. The mount does well to properly grip the shaft and the flywheel with any issues.

Cause: The hole in the mount was too shallow and the shaft sticks up from the flywheel. Additionally the mount has a quarter inch lip on the bottom that should be changed.

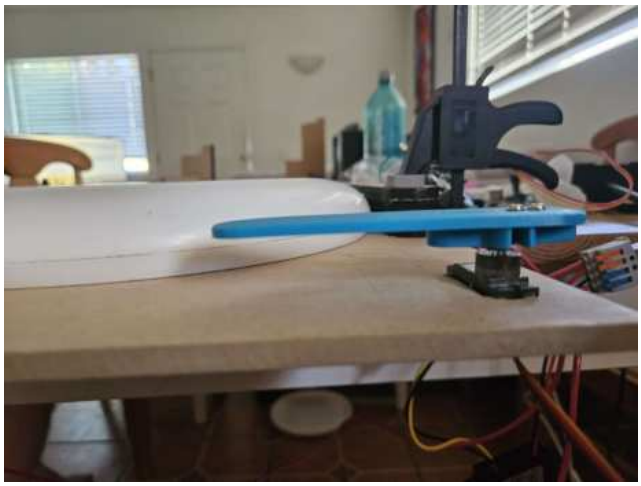
Analysis (Solution): Drill the full length of the shaft into the shaft mount and use the threaded aspect to secure it so the flywheel actually sits close to the track

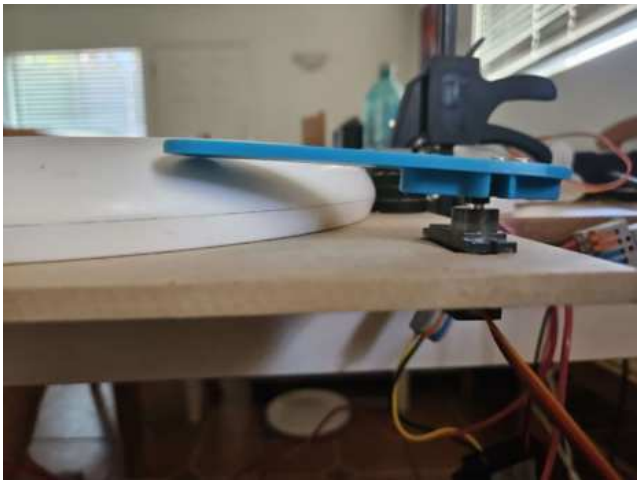
RCA2:

Root Problem: Servo arm doesn't push the frisbee

Cause: the servo arm sits above the frisbee causing it to slide up the frisbee rather than pushing it

Analysis (Solution): either mount the servo lower down, or add a little piece at the bottom of the 3d printed arm to have it push the frisbee



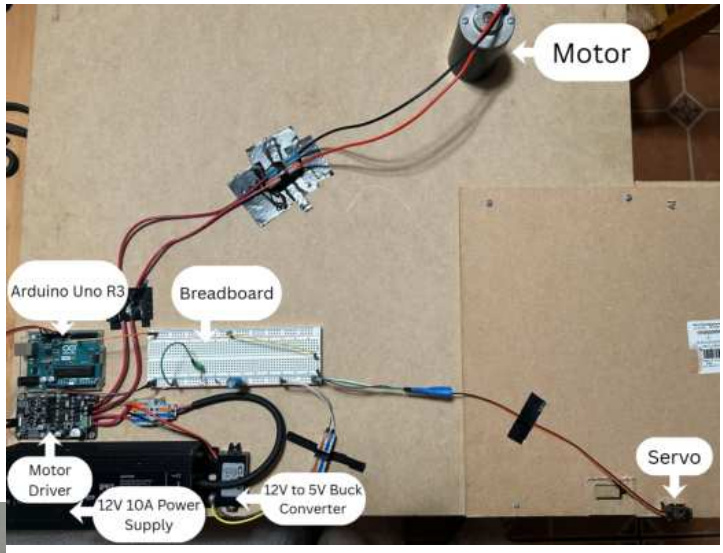


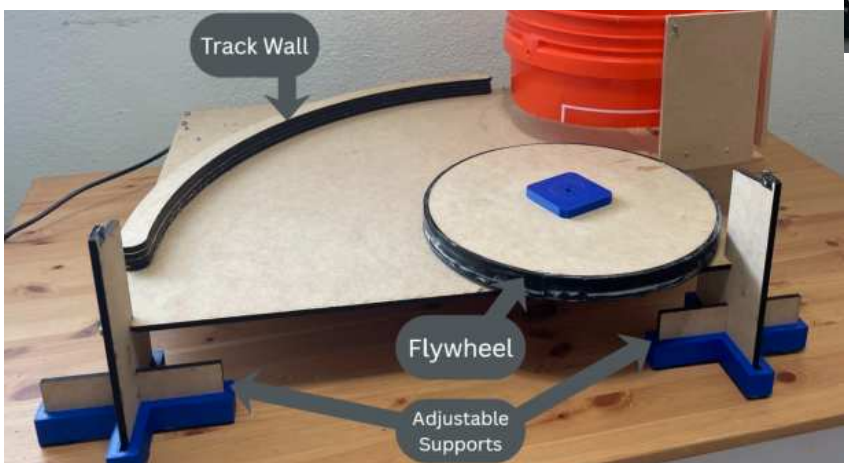
RCA3:
Problem: Track overall works great, snugly fits motor, servo, and wall piece. However, the legs are a little wobbly but seem to work well and have adjustable loft.
Cause: There is no glue or slot that constrains the two leg pieces from sliding on each other.
Solution: I will start with wood glue, see if that is strong enough, and if not cut a small notch for the pieces to fit together.

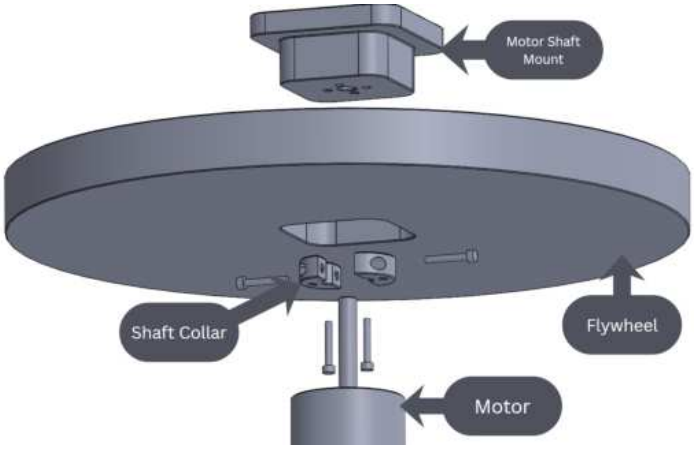
Final Build:

1.









2.
Our launcher features a 30 cm diameter, 1.4 kg laser-cut wood flywheel. A rubber strip is glued onto the outer rim of the flywheel to improve grip. A laser cut curved track sits roughly one Frisbee diameter away from the flywheel to ensure a snug fit. Based on required rotational energy calculations, we scaled our flywheel to have enough inertia to launch the disc 15 m without a significant RPM drop. Additionally, we carefully offset the track from the flywheel to maximize normal forces and thus frictional acceleration of the disc, without deforming it.

The motor and the arduino run at different voltages so a buck converter is necessary for all components to be powered by the same power supply. The arduino communicates with the driver to ramp the motor up to speed without spiking the current and causes the servo to push the Frisbee into the flywheel at a regular interval once the motor reaches top speed. The motor is rated at 0.098 Nm and has a total spinup time of 45 seconds. The initial goal was a 10 second spinup time, but we were unable to acquire a higher torque motor due to shipping error.

The launcher is able to consistently launch Frisbees in the same direction and at the same height across short distances.

Section III - Other Evidence

1. Code to Calculate flywheel size

$v_w = v_d = v \quad \rightarrow \quad R_w\omega_w = R_d\omega_d \quad \rightarrow \quad v = R_w\omega_w \quad \rightarrow \quad \frac{\omega_d}{\omega_w} = \frac{R_w}{R_d}$

Breaking down the energy stored in the flywheel:

$E_w = \frac{1}{2} I_w \omega_w^{\ 2}$

Flywheel inertia for a disc (replace with actual inertia value):

$I_w = \frac{1}{2} m_w R_w^{\ 2}$

$E_w = \frac{1}{4} m_w R_w^{\ 2} \omega_w^{\ 2} \quad \rightarrow \quad E_w = \frac{1}{4} m_w v^2$

Energy of the frisbee:

$E_d = \frac{1}{2} m_d v^2 + \frac{1}{2} I_d \omega_d^{\ 2} \quad \rightarrow \quad I_d = \frac{1}{2} m_d R_d^{\ 2} \quad \rightarrow \quad E_d = \frac{3}{4} m_d v^2$

Required RPM:

$RPM = \frac{v}{2\pi R_w} * 60$

Assuming we want no more than a 20% Energy drop:

$E_w > 5E_d \quad \rightarrow \quad m_w > 15m_d$

$\tau = I_w \alpha \quad \rightarrow \quad \alpha = \frac{\Delta\omega}{t_{spinup}} \quad \rightarrow \quad \tau = I_w * \frac{\Delta\omega}{t_{spinup}}$

Assuming a spinup time of 10 seconds:

$\tau = I_w * \frac{\Delta\omega}{10} \quad \rightarrow \quad \tau = \frac{1}{2} m_w R_w^{\ 2} * \frac{\Delta\omega}{10}$

Section IV - Reflect

1. One thing that we were surprised by was how much small alignment and mounting issues affected overall performance. Additionally, creating useful prototypes to answer design questions was more challenging than we expected.
2. We succeeded in proving the launch concept works but did not fully refine the loading and mounting systems.
3. Yes, the prototype showed that the concept is viable and can be improved with further development.
4. We would improve the flywheel mount, servo mechanism, and structural stability. Increasing the size of the flywheel to transfer more energy into the Frisbee and using a uniform grippy material on the track wall should improve the launch distance.
5. We would start prototyping earlier to allow more time for testing and design iterations. Developing different testing methods that could provide us with helpful information without committing to full scale builds would improve the design and be more resource efficient.